Lithium Cell + DC Boost

for motor/buzzer/ 5V I2C / LEDs ...

if you put the battery in the wrong orientation, +5V the HW-668 will DIE !

U2 HW-668 (VIN, OUT, GND, GND) — DC BOOST — use SX1308 inside

U1 Lithium Holder 18650 (V_BAT, GND)

J1 Conn_01x02 — conn to Lolin32

ESP32 Devboard (Lolin32 Lite) - U6

2.4Ghz Ant

ESP32-D0WD(-V3) CH340C USB/UART TP4054 charger (500mA)

Lolin32 Lite (BAT, USB-C)

RE_Right_A, RE_Right_B, RE_Left_A, RE_Left_B, IR_analog, PCA9685_OE, BUZZER, IR_MODULATION, WS2812_DIN, A1, RE_Left_btn

IR_S0, SDA, IR_S1, SCL, IR_S2, B2, B1, A2, RE_Right_btn

dont press the rotary encoder during boot...

Rotary Encoder (HW-04)

U9: RE_Left_A, RE_Left_B → RE_Left_btn

U8: RE_Right_A, RE_Right_B → RE_Right_btn

the HW-040 has the pull up resistor for the rotary part but you need to use the MCU internal pullup to use the switch

C5, C6 10 nF

U13, U22 VCC, VEE, GND

IR_analog, IR_S0, IR_S1, IR_S2

IR_REC_0 … IR_REC_7

IR_TRA_0 … IR_TRA_7

IR_MODULATION

there is ~70 Ohm trough the chip. beware

Vee is for negative voltage but the lowest here is GND

DC motor Driver

if the board change and we have more than 5V, nsleep require a resistor

U4 DRV8833: AIN1, AIN2, VINT, GND, VM, VCP, BIN2, BIN1, nSLEEP, AOUT1, AISEN, AOUT2, BOUT2, BISEN, BOUT1, nFAULT

C1 10 uF, C2 2.2uF, C3 10 nF

R1 0R, R2 0R

just short theses pad unless you know why you need a resistor in which case the pad is made for U

who read error message anyway ? nFault wont be needed then

DRV8833 breakout board (black / HW-627): IN4/BIN2, IN3/BIN1, GND, VCC, IN2/AIN2, IN1/AIN1, SLEEP, AOUT1/OUT1, AOUT2/OUT2, BOUT1/OUT3, BOUT2/OUT4, FAULT

J5 Conn_01x02, J6 Conn_01x02

U5

the breakout board can be mounted or the individual component can be soldered but not both at the same time.

Buzzer

R7 R, BZ1 Buzzer, Q3 2N7000

BUZZER

pull down seems overkill here, if needed, just do it in software

no logic level, worst case scenario, the first LED will glitch and act as a level shifter

WS2812 RBG addressable LED

WS2812_DIN

U7, U10, U11, U12 (DIN, VDD, DOUT, VSS, RGB)

J10 5V D GND Conn_01x03

theses connector allow to cut the PCB and place the IR sensor where you want using a bit of cable

R8–R15 IR_REC_0 … IR_REC_7

R16–R23 IR_TRA_0 … IR_TRA_7

U14–U21 ITR9909

Q4–Q11 2N7000

J11–J26 Conn_01x04

Servo Driver

SolderJumper_2_Bridged JP1

the capa should be as big as you can to absorb the inrush current of the servo

C4 10 uF

U3 PCA9685PW: SCL, SDA, EXTCLK, OE, A0–A5, VDD, VSS, LED0–LED15

PCA9685_OE

Address : 1111 010x

J2, J3, J4 Conn_01x16

if you require multiple servo and need to have torque, the current can be very high, way more than what the DC boost can provide you will need to adapt the 5V power supply

this is why you have a solder jumper to wire you own supply as needed

5V I2C connector

R3 R, R6 R, Q1 2N7000

also serve as I2C pullup

R4 R, R5 R, Q2 2N7000

SDA, SCL

if the layout dont match your board, you can swap the connection using theses pads

JP2–JP5 SolderJumper_2_Bridged

J9 SDA SCL 5V GND Conn_01x04

made to match something like this by default
https://de.aliexpress.com/item/1005008640132638.html

3V3 I2C connector (QWIIC)

J7, J8 SCL SDA +3V3 GND Conn_01x04

Yellow = SCL
Blue = SDA
Red = 3V3
Black = GND

https://www.sparkfun.com/qwiic#products

Nathann Morand
Robopoly

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Title: Kit Prisme

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